# Dr. Bhaskar Goswami

## M.Sc., Ph.D., FRAS

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## Short Bio-

**I am a physicist with a deep curiosity about the fundamental laws of nature and the physics of the universe. My research focuses on the light-scattering properties of cosmic dust and their role in shaping the observable universe. I investigate how tiny dust particles, such as those found in comets, asteroids, and interstellar clouds, influence planetary systems and contribute to the evolution of life on Earth. Using computational modeling and polarimetric techniques, I study the scattering behavior of dust in comets and asteroids, particularly their activity within the solar system as they approach the Sun, as well as the polarization properties of dust in dark clouds and star clusters.**

## Profiles

**https://rb.gy/exkzpa**

**https://shorturl.at/FQb25**

**https://shorturl.at/btq**

## Interests

- Observational Astronomy
- Dust Modelling
- Polarimetry of comets and Asteroids
- Artificial Intelligence
- Cloud Computing

## Education

### Postgraduate Education

**Sept. 2020-Dec. 2025:** Ph.D. in Physics

**Assam University, Silchar, India**

**Thesis Title:** Study of optical properties of cosmic dust through light scattering

**Supervisor:** Prof. Himadri Sekhar Das.

**Aug. 2017-Sept. 2019:** M.Sc. in Physics

**University of Science and Technology, Meghalaya, India**

**Dissertation Title:** Small Ammeture Radio Telescope (SAMRAT)

**Supervisor:** Dr. Nitu Borgohain.

**CGPA:** 8.1 (81%)

### Undergraduate Education

**Aug. 2014-Sept. 2017**: B.Sc. in Physics

**Dhing College, Gauhati University, India**

**Dissertation Title:** ELECTRO URJA SAVER (EUS)

**Supervisor:** Dr. Nandan Barthakur.

**CGPA:** 7.3 (66.52%)

### Research schools/research Projects

1. **21-27 November 2022:** IFAS (Indo-French Astronomy School) at the Inter-University Centre for Astronomy and Astrophysics (IUCAA).

2. **8-25 March 2019:** 'SITARE' (Southampton-IUCAA training for Astrophysics Research and Education) at the University of Southampton, U.K. approved grant funded by STFC (The Science and Technology Facilities Council) under the GCRF (Global Challenges Research Fund) initiative (Grant Ref: ST/R002797/1).

## Skills

### Programming

- Python
- Fortran
- Latex
- C++

### Tools

- IRAF
- PyRAF
- HTML
- UNIX

### Languages

- English
- Assamese
- Bengali
- Hindi

# List of Publications

## Published papers

1. **Goswami, B.**, Das, H. S., Devi, N., & Halder, P. (2024). **Influence of porosity on the Umov effect in silicate and organic refractory aggregates.** ***Journal of Quantitative Spectroscopy and Radiative Transfer****, 323, 109019.*

2. **Goswami, B.**, & Das, H. S. (2023). **A study of the correlation between polarization maximum to minimum ratio and scattering parameters in a spheroid dust model.** ***Journal of Quantitative Spectroscopy and Radiative Transfer****, 309, 108689.*

3. **Goswami, B.**, & Das, H. S. (2022). **Study of the Umov effect with carbon aggregates.** ***Chemical Journal of Karimganj College,*** *6(1), 35–39.*

## Co-authored papers

4. Choudhury, G., **Goswami, B.**, Das, H., Medhi, B., & Pandey, J. (2024). **Probing the magnetic field and dust grain properties of two dark clouds L1495 and L1498 through photopolarimetry.** ***Monthly Notices of the Royal Astronomical Society****, 528(4), 7156–7164. https://doi.org/10.1093/mnras/stae100*

5. Devi, N. N., Das, H. S., Suklabaidya, A., & **Goswami, B.** (2025). **The Umov effect in polydisperse dust aggregates.** ***Journal of Quantitative Spectroscopy and Radiative Transfer. Advance online publication****. https://doi.org/10.1016/j.jqsrt.2025.xxxxx*

6. Prasad, B., Das, H. S., Choudhury, G. B., Pandey, J. C., Medhi, B. J., Paul, R. S., Paul, D., Barman, B., Nath Mazumdar, S., Devi, N. N., & Goswami, B. (2025). Phase-Polarisation Analysis of Asteroids 13 Egeria and 511 Davida: New Low Phase Angle Observations and C-Type Comparisons. Accepted in Solar System Research journal (In Press)

## Book Chapters

1. **Goswami, B.**, & Saikia, K. (2017, October 14). **An assessment of noise pollution on the east side of Guwahati city. In** ***Proceedings of the UGC Sponsored National Seminar on Issues of Traffic Noise Pollution on Environment and Control Measures in Northeast India*** *(ISBN: 978-81-202-8809-8).*

2. Prasad, B., Das, H. S., & **Goswami, B.** (2025, September). **Light scattering behavior of near-Earth asteroids: A high-phase-angle polarimetric study. In** ***Contemporary Explorations in Multidisciplinary Sciences*** *(ISBN: 978-93-48301-03-1).*

## List of conferences

### International Conferences

1. **Goswami, B.**, & Das, H. S. (2024, February 29–March 2). **Study of correlation between polarization maximum and geometric albedo using aggregate dust model.** ***Frontiers in Pure and Applied Physics (ICFPAP-2024), Department of Physics,*** *University of Science and Technology, Meghalaya, India.*

### National Conferences

1. **Goswami, B.**, Prasad, B., & Das, H. S. (2024, October 23–25). **Polarimetric study of asteroids 4 Vesta and 10 Hygiea.** ***10th North East Meet of Astronomers****, Department of Physics, Tezpur University, Assam, India.*

2. **Goswami, B.**, & Das, H. S. (2024, February 17–18). **Study of the Umov effect using carbon and silicate aggregates.** ***Recent Developments in Science and Technology (NSRDST-2024)****, Department of Chemistry, Karimganj College (under the aegis of IQAC), Assam, India.*

3. **Goswami, B.**, & Das, H. S. (2023, November 20–23). **Study of the Umov effect in carbon and silicate aggregates.** ***9th North East Meet of Astronomers,*** *Department of Physics, Mizoram University, Mizoram, India.*

4. **Goswami, B.**, & Das, H. S. (2022, March 25–29). **Modeling of polarization properties of spheroidal particles: Correlation between polarization maximum and minimum.** ***40th Meeting of the Astronomical Society of India,*** *Indian Institute of Technology Roorkee & ARIES, Nainital, India.*

## LEADERSHIP AND ACTIVITIES

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| **28 March-9 November 2025** | **:** | **President**, Ad-hoc committee, Assam University Students' Union. |
| **8-10 March 2025** | **:** | **Scholar Organizing Secretary**, International Young Researchers' Conclave and Rural Science Congress. |
| **1 December 2023 – 15 March 2025** | : | **General Secretary**, Assam University Research Scholars' Forum (AURSF). |
| **08-09 March 2023** | **:** | **General Secretary**, SCIENTIA_4.0 (General Science Fest of Assam University Silchar). |
| **1 December 2022 – 30 November 2024** | **:** | **Member**, Assam University Research Scholars' Forum. |

## Associated Societies

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| **16 December 2019 onwards** | **:** | Elected as the lifetime **Fellow of the Royal Astronomical Society** (entitled to write **FRAS**). (**Membership Number: 10363**) |